

GEO - HUBBLE MATHS

GEO - Geometric Resolution of the Hubble Tension

Formal demonstration of 12-digit reconstruction and strong prediction condition

0. Scope: Separating two levels: **Level A (Reconstructive Audit)** and **Level B (Strong Prediction)**.
Exact reconstruction serves as technical evidence that the tension admits a compact geometric form.

Magnitude	Value
H0 base Planck/LCDM	67.40 km/s/Mpc
H0 local - SHOES	73.04 km/s/Mpc
Relative Difference	8.36795252%

2. GEO Reading Law: $T = A + B + L$. For Hubble resolution: $\eta = 0.6$; $A = 0.6$; $B = 0.4$; $fc = \sqrt{\eta} \approx 0.774597$; $R = \eta^{(1/3)} \approx 0.843433$.

4. GEO-Lens Operator: $H0_GEO = H0_base * (1 + \alpha * O_GEO)$. Where $\alpha = \Delta H / O_GEO$.

GEO Operator	O GEO	Required Alpha	Reconstructed H0
delta_R	0.156567334698	0.534463497024	73.040000000000
B/A	0.666666666667	0.125519287834	73.040000000000
B = 1 - η	0.400000000000	0.209198813056	73.040000000000

6. Strong Prediction: Alpha must be derived from internal GEO geometry (Node 10, 45.01° angle) before comparing against SHOES.